Assignment 3 Report | 1905074

How to run code from 1905074.ipynb:

1. Import everything from **Imports** section
2. Load dataset from **Dataset Loading** section
3. Run **Neural Network** section
4. Load the best model from **Load Model** section
5. Test Model from **Testing Model**

Training model with architecture [784, 128, 10] and learning rate 0.005

Epoch 1/10 - Loss: 0.0444, Accuracy: 0.8419, Val Loss: 0.0457, Val Accuracy: 0.8381, Val F1: 0.8379

Epoch 2/10 - Loss: 0.0400, Accuracy: 0.8545, Val Loss: 0.0420, Val Accuracy: 0.8489, Val F1: 0.8474

Epoch 3/10 - Loss: 0.0385, Accuracy: 0.8606, Val Loss: 0.0407, Val Accuracy: 0.8548, Val F1: 0.8529

Epoch 4/10 - Loss: 0.0368, Accuracy: 0.8661, Val Loss: 0.0393, Val Accuracy: 0.8586, Val F1: 0.8583

Epoch 5/10 - Loss: 0.0367, Accuracy: 0.8662, Val Loss: 0.0401, Val Accuracy: 0.8565, Val F1: 0.8550

Epoch 6/10 - Loss: 0.0345, Accuracy: 0.8738, Val Loss: 0.0378, Val Accuracy: 0.8647, Val F1: 0.8635

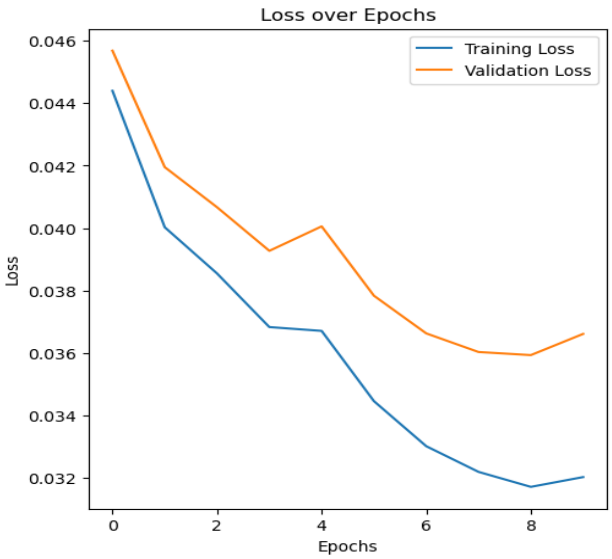
Epoch 7/10 - Loss: 0.0330, Accuracy: 0.8809, Val Loss: 0.0366, Val Accuracy: 0.8707, Val F1: 0.8702

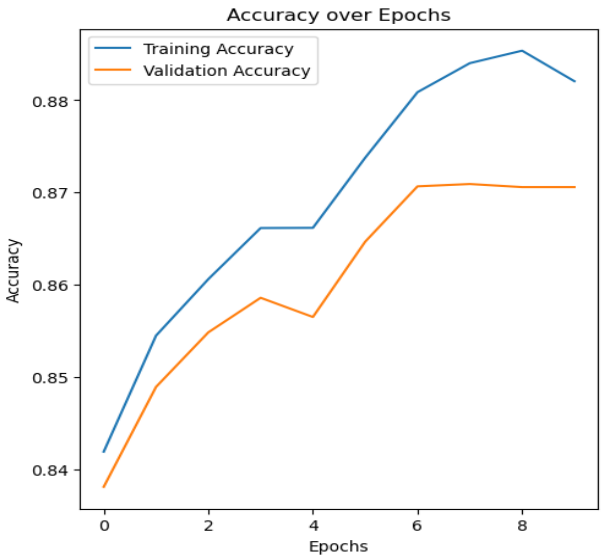
Epoch 8/10 - Loss: 0.0322, Accuracy: 0.8840, Val Loss: 0.0360, Val Accuracy: 0.8709, Val F1: 0.8704

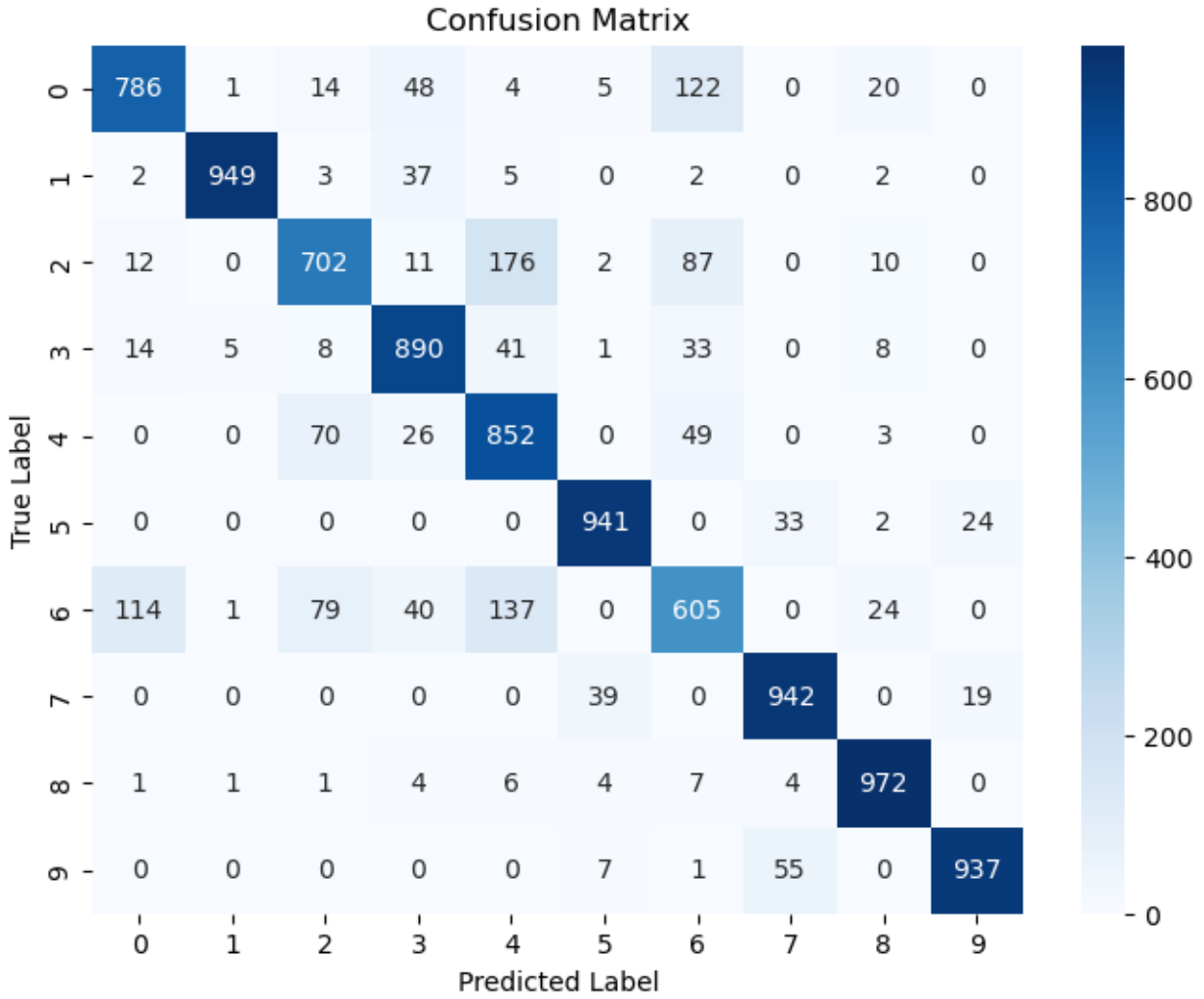
Epoch 9/10 - Loss: 0.0317, Accuracy: 0.8854, Val Loss: 0.0359, Val Accuracy: 0.8706, Val F1: 0.8705

Epoch 10/10 - Loss: 0.0320, Accuracy: 0.8821, Val Loss: 0.0366, Val Accuracy: 0.8706, Val F1: 0.8703

Validation Loss: 0.0366, Validation Accuracy: 0.8706, Validation F1: 0.8703


Loss over Epochs
Training Loss
Validation Loss
Loss
Epochs
0.046
0.044
0.042
0.040
0.038
0.036
0.034
0.032
0
2
4
6
8


Accuracy over Epochs
Training Accuracy
Validation Accuracy
Accuracy
Epochs
0.88
0.87
0.86
0.85
0.84
0
2
4
6
8


Confusion Matrix
True Label
Predicted Label
0 786 1 14 48 4 5 122 0 20 0
1 2 949 3 37 5 0 2 0 2 0
2 12 0 702 11 176 2 87 0 10 0
3 14 5 8 890 41 1 33 0 8 0
4 0 0 70 26 852 0 49 0 3 0
5 0 0 0 0 0 941 0 33 2 24
6 114 1 79 40 137 0 605 0 24 0
7 0 0 0 0 0 39 0 942 0 19
8 1 1 1 4 6 4 7 4 972 0
9 0 0 0 0 0 7 1 55 0 937
0 1 2 3 4 5 6 7 8 9
800
600
400
200
0

Training model with architecture [784, 128, 10] and learning rate 0.001

Epoch 1/10 - Loss: 0.0656, Accuracy: 0.7623, Val Loss: 0.0668, Val Accuracy: 0.7628, Val F1: 0.7609

Epoch 2/10 - Loss: 0.0527, Accuracy: 0.8141, Val Loss: 0.0537, Val Accuracy: 0.8100, Val F1: 0.8067

Epoch 3/10 - Loss: 0.0463, Accuracy: 0.8328, Val Loss: 0.0471, Val Accuracy: 0.8285, Val F1: 0.8249

Epoch 4/10 - Loss: 0.0431, Accuracy: 0.8461, Val Loss: 0.0446, Val Accuracy: 0.8389, Val F1: 0.8360

Epoch 5/10 - Loss: 0.0403, Accuracy: 0.8558, Val Loss: 0.0419, Val Accuracy: 0.8484, Val F1: 0.8469

Epoch 6/10 - Loss: 0.0396, Accuracy: 0.8571, Val Loss: 0.0414, Val Accuracy: 0.8503, Val F1: 0.8480

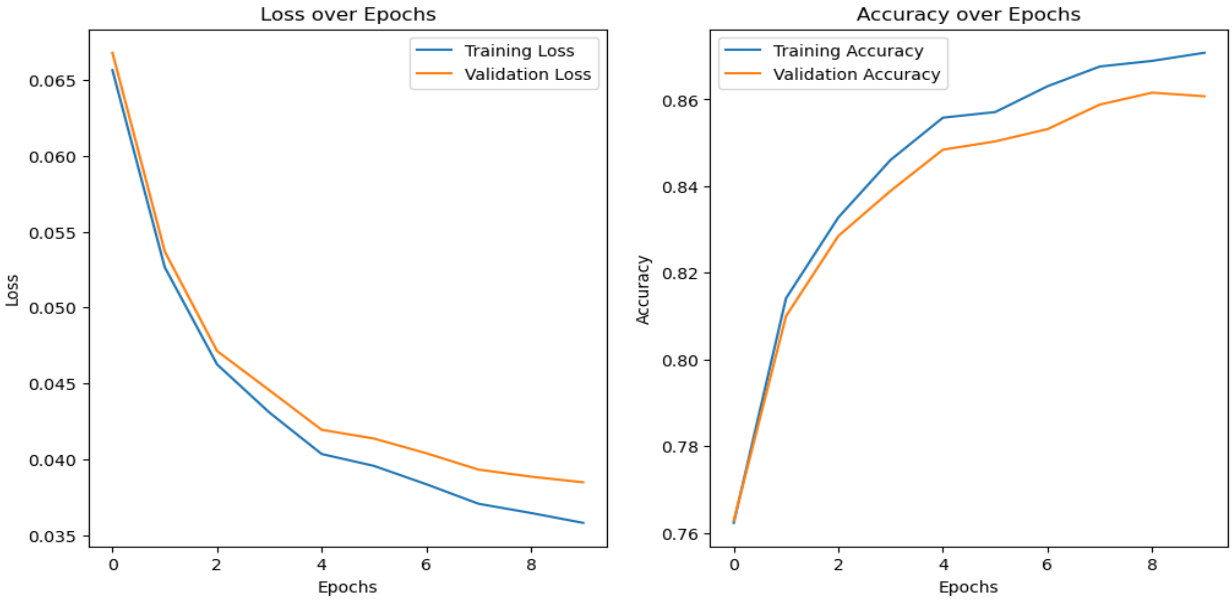
Epoch 7/10 - Loss: 0.0384, Accuracy: 0.8631, Val Loss: 0.0404, Val Accuracy: 0.8532, Val F1: 0.8513

Epoch 8/10 - Loss: 0.0371, Accuracy: 0.8676, Val Loss: 0.0393, Val Accuracy: 0.8588, Val F1: 0.8574

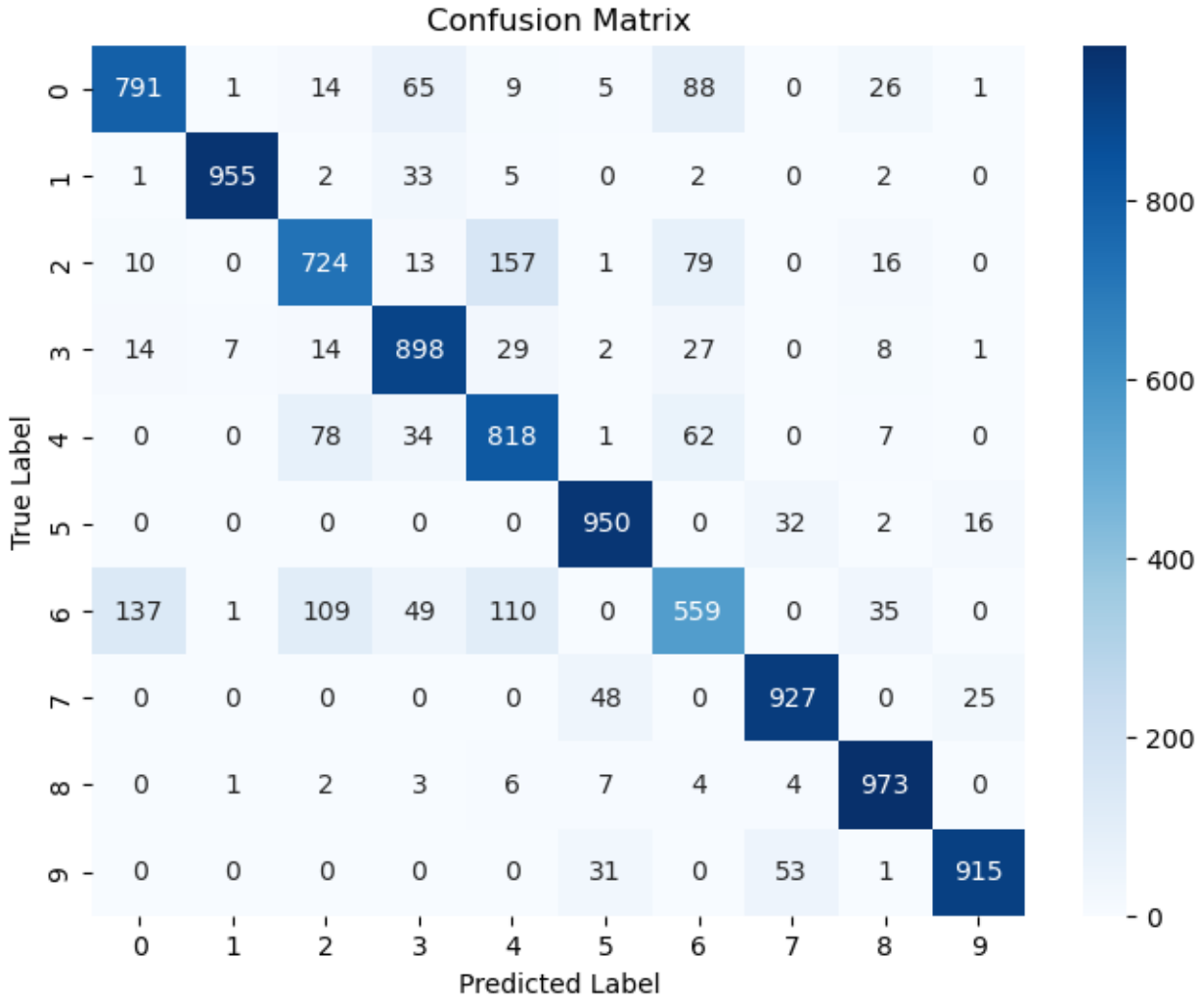
Epoch 9/10 - Loss: 0.0365, Accuracy: 0.8689, Val Loss: 0.0389, Val Accuracy: 0.8616, Val F1: 0.8603

Epoch 10/10 - Loss: 0.0358, Accuracy: 0.8708, Val Loss: 0.0385, Val Accuracy: 0.8608, Val F1: 0.8592

Validation Loss: 0.0385, Validation Accuracy: 0.8608, Validation F1: 0.8592


Loss over Epochs
Training Loss
Validation Loss
Loss
Epochs
0.065
0.060
0.055
0.050
0.045
0.040
0.035
0
2
4
6
8

Accuracy over Epochs
Training Accuracy
Validation Accuracy
Accuracy
Epochs
0.86
0.84
0.82
0.80
0.78
0.76
0
2
4
6
8


Confusion Matrix
True Label
Predicted Label
0 791 1 14 65 9 5 88 0 26 1
1 1 955 2 33 5 0 2 0 2 0
2 10 0 724 13 157 1 79 0 16 0
3 14 7 14 898 29 2 27 0 8 1
4 0 0 78 34 818 1 62 0 7 0
5 0 0 0 0 0 950 0 32 2 16
6 137 1 109 49 110 0 559 0 35 0
7 0 0 0 0 0 48 0 927 0 25
8 0 1 2 3 6 7 4 4 973 0
9 0 0 0 0 0 31 0 53 1 915
0 1 2 3 4 5 6 7 8 9
800
600
400
200
0

Training model with architecture [784, 128, 10] and learning rate 0.0005

Epoch 1/10 - Loss: 0.0860, Accuracy: 0.7036, Val Loss: 0.0873, Val Accuracy: 0.7026, Val F1: 0.6977

Epoch 2/10 - Loss: 0.0690, Accuracy: 0.7518, Val Loss: 0.0701, Val Accuracy: 0.7506, Val F1: 0.7376

Epoch 3/10 - Loss: 0.0587, Accuracy: 0.7898, Val Loss: 0.0599, Val Accuracy: 0.7857, Val F1: 0.7807

Epoch 4/10 - Loss: 0.0537, Accuracy: 0.8040, Val Loss: 0.0549, Val Accuracy: 0.8006, Val F1: 0.7987

Epoch 5/10 - Loss: 0.0506, Accuracy: 0.8178, Val Loss: 0.0517, Val Accuracy: 0.8127, Val F1: 0.8101

Epoch 6/10 - Loss: 0.0476, Accuracy: 0.8266, Val Loss: 0.0489, Val Accuracy: 0.8207, Val F1: 0.8190

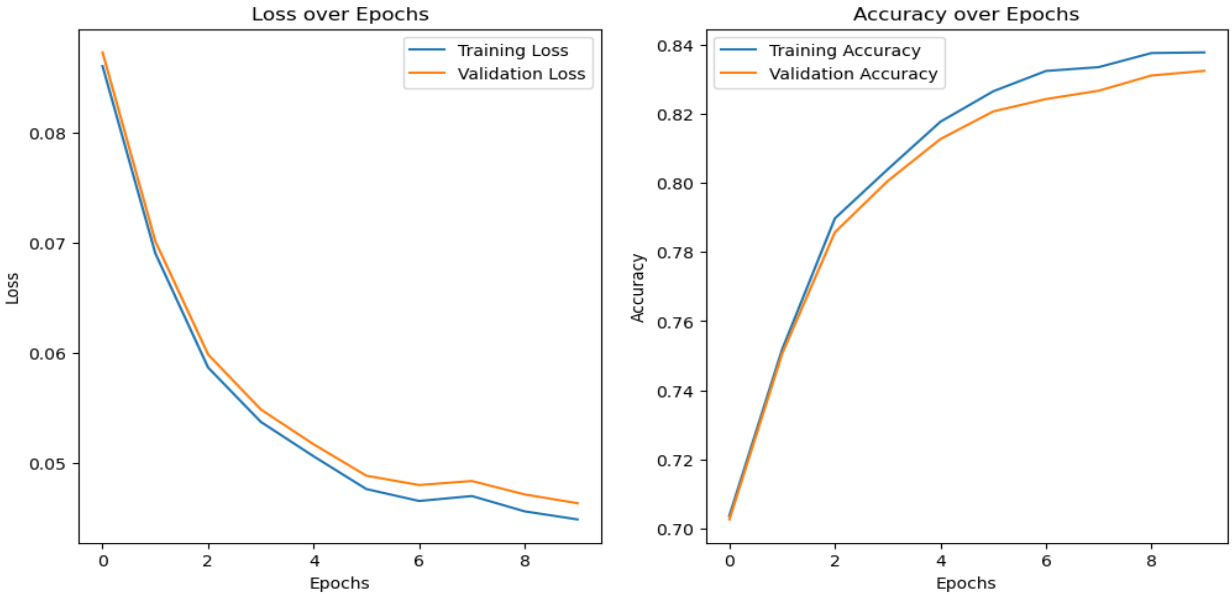
Epoch 7/10 - Loss: 0.0466, Accuracy: 0.8325, Val Loss: 0.0480, Val Accuracy: 0.8243, Val F1: 0.8221

Epoch 8/10 - Loss: 0.0470, Accuracy: 0.8336, Val Loss: 0.0484, Val Accuracy: 0.8267, Val F1: 0.8256

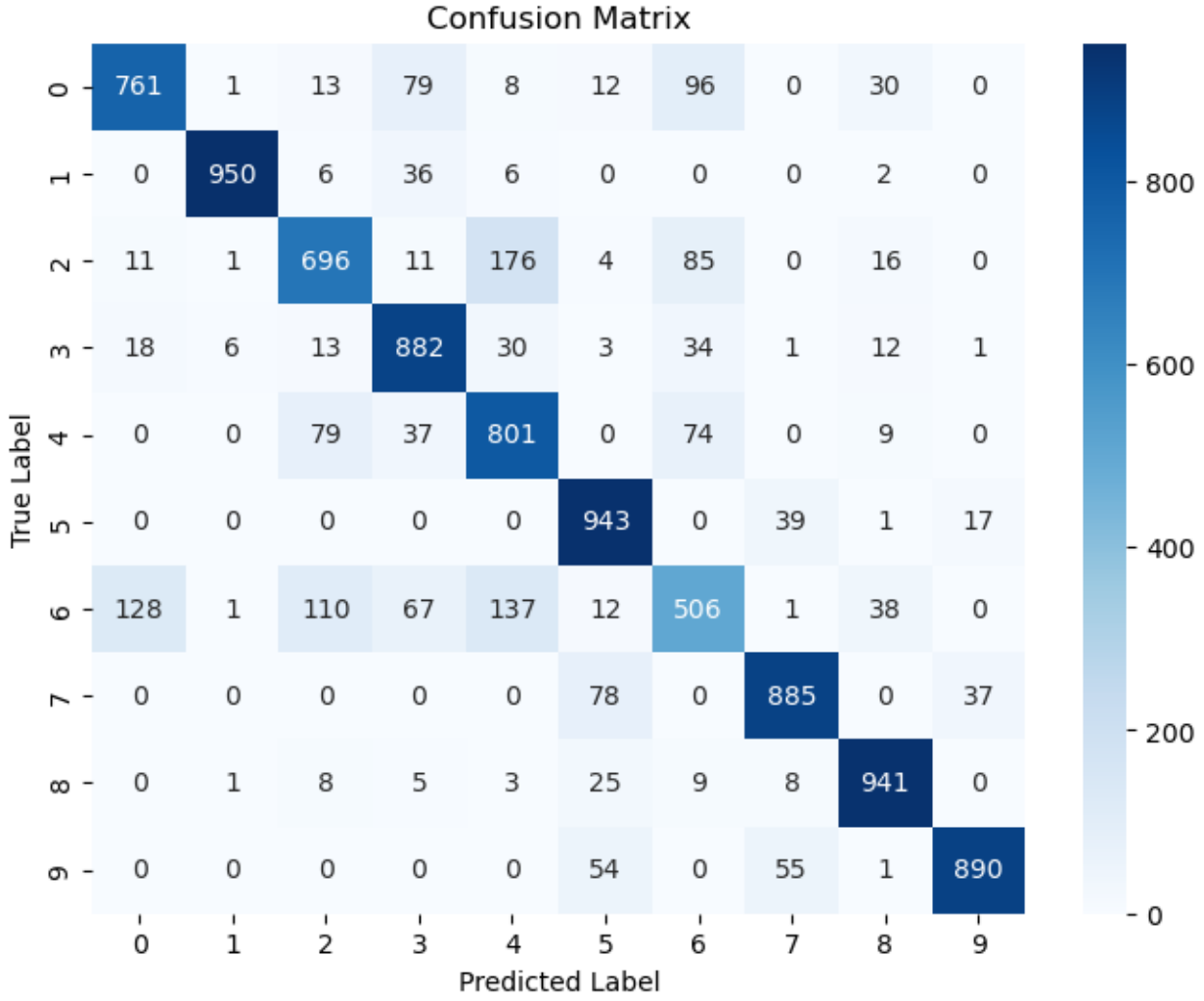
Epoch 9/10 - Loss: 0.0456, Accuracy: 0.8377, Val Loss: 0.0472, Val Accuracy: 0.8312, Val F1: 0.8287

Epoch 10/10 - Loss: 0.0449, Accuracy: 0.8379, Val Loss: 0.0464, Val Accuracy: 0.8325, Val F1: 0.8308

Validation Loss: 0.0464, Validation Accuracy: 0.8325, Validation F1: 0.8308


Loss over Epochs
Training Loss
Validation Loss
Loss
Epochs
0.08
0.07
0.06
0.05
0
2
4
6
8

Accuracy over Epochs
Training Accuracy
Validation Accuracy
Accuracy
Epochs
0.84
0.82
0.80
0.78
0.76
0.74
0.72
0.70
0
2
4
6
8


Confusion Matrix
True Label
Predicted Label
0 761 1 13 79 8 12 96 0 30 0
1 0 950 6 36 6 0 0 0 2 0
2 11 1 696 11 176 4 85 0 16 0
3 18 6 13 882 30 3 34 1 12 1
4 0 0 79 37 801 0 74 0 9 0
5 0 0 0 0 0 943 0 39 1 17
6 128 1 110 67 137 12 506 1 38 0
7 0 0 0 0 0 78 0 885 0 37
8 0 1 8 5 3 25 9 8 941 0
9 0 0 0 0 0 54 0 55 1 890
0 1 2 3 4 5 6 7 8 9
800
600
400
200
0

Training model with architecture [784, 64, 10] and learning rate 0.005

Epoch 1/10 - Loss: 0.0496, Accuracy: 0.8253, Val Loss: 0.0508, Val Accuracy: 0.8201, Val F1: 0.8179

Epoch 2/10 - Loss: 0.0432, Accuracy: 0.8444, Val Loss: 0.0447, Val Accuracy: 0.8377, Val F1: 0.8362

Epoch 3/10 - Loss: 0.0411, Accuracy: 0.8491, Val Loss: 0.0430, Val Accuracy: 0.8447, Val F1: 0.8422

Epoch 4/10 - Loss: 0.0402, Accuracy: 0.8557, Val Loss: 0.0423, Val Accuracy: 0.8482, Val F1: 0.8468

Epoch 5/10 - Loss: 0.0388, Accuracy: 0.8604, Val Loss: 0.0411, Val Accuracy: 0.8519, Val F1: 0.8495

Epoch 6/10 - Loss: 0.0381, Accuracy: 0.8624, Val Loss: 0.0408, Val Accuracy: 0.8542, Val F1: 0.8528

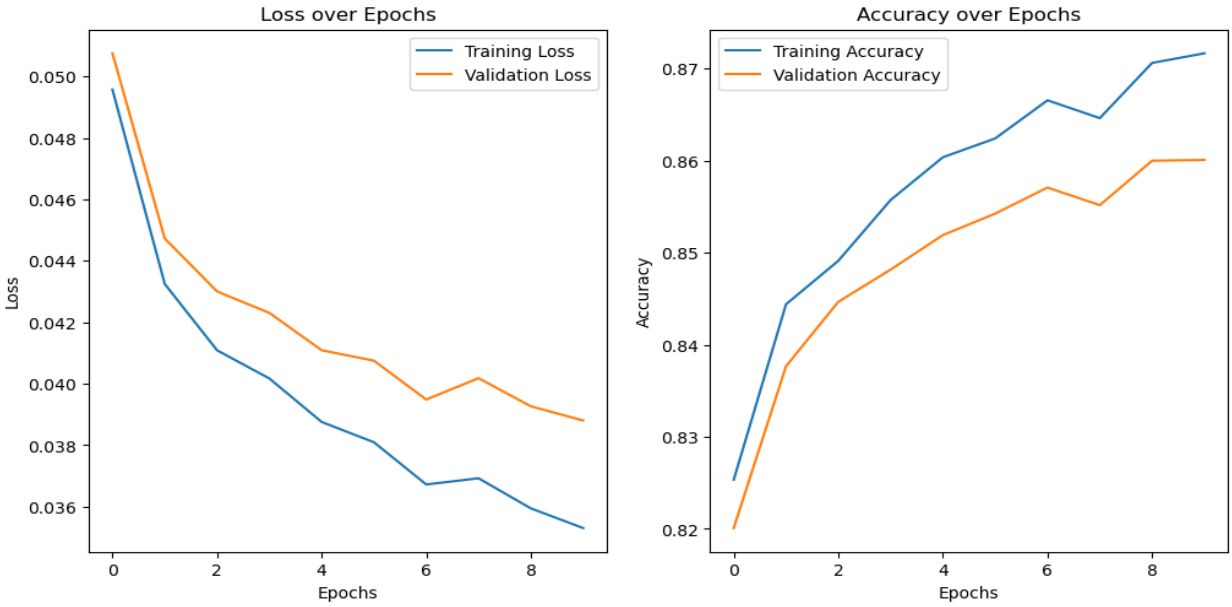
Epoch 7/10 - Loss: 0.0367, Accuracy: 0.8666, Val Loss: 0.0395, Val Accuracy: 0.8571, Val F1: 0.8559

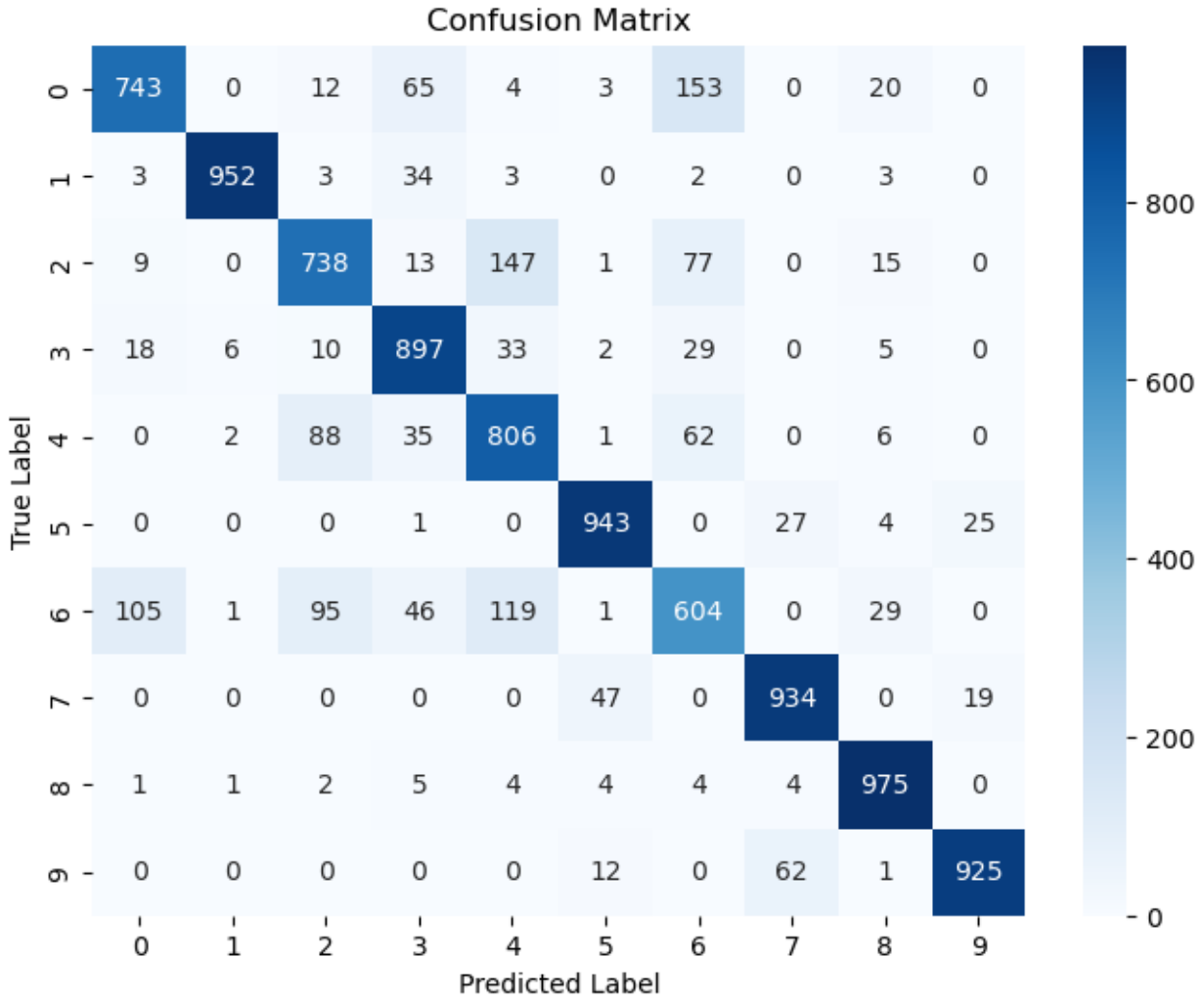
Epoch 8/10 - Loss: 0.0369, Accuracy: 0.8646, Val Loss: 0.0402, Val Accuracy: 0.8552, Val F1: 0.8511

Epoch 9/10 - Loss: 0.0359, Accuracy: 0.8706, Val Loss: 0.0393, Val Accuracy: 0.8600, Val F1: 0.8582

Epoch 10/10 - Loss: 0.0353, Accuracy: 0.8717, Val Loss: 0.0388, Val Accuracy: 0.8601, Val F1: 0.8597

Validation Loss: 0.0388, Validation Accuracy: 0.8601, Validation F1: 0.8597


Loss over Epochs
Training Loss
Validation Loss
Loss
Epochs
Accuracy over Epochs
Training Accuracy
Validation Accuracy
Accuracy
Epochs


Confusion Matrix
True Label
Predicted Label
743 0 12 65 4 3 153 0 20 0
3 952 3 34 3 0 2 0 3 0
9 0 738 13 147 1 77 0 15 0
18 6 10 897 33 2 29 0 5 0
0 2 88 35 806 1 62 0 6 0
0 0 0 1 0 943 0 27 4 25
105 1 95 46 119 1 604 0 29 0
0 0 0 0 0 47 0 934 0 19
1 1 2 5 4 4 4 4 975 0
0 0 0 0 0 12 0 62 1 925

Training model with architecture [784, 64, 10] and learning rate 0.001

Epoch 1/10 - Loss: 0.0697, Accuracy: 0.7592, Val Loss: 0.0706, Val Accuracy: 0.7572, Val F1: 0.7486

Epoch 2/10 - Loss: 0.0563, Accuracy: 0.8015, Val Loss: 0.0575, Val Accuracy: 0.7947, Val F1: 0.7893

Epoch 3/10 - Loss: 0.0509, Accuracy: 0.8226, Val Loss: 0.0520, Val Accuracy: 0.8166, Val F1: 0.8143

Epoch 4/10 - Loss: 0.0467, Accuracy: 0.8367, Val Loss: 0.0480, Val Accuracy: 0.8311, Val F1: 0.8282

Epoch 5/10 - Loss: 0.0437, Accuracy: 0.8464, Val Loss: 0.0451, Val Accuracy: 0.8395, Val F1: 0.8370

Epoch 6/10 - Loss: 0.0422, Accuracy: 0.8509, Val Loss: 0.0438, Val Accuracy: 0.8442, Val F1: 0.8421

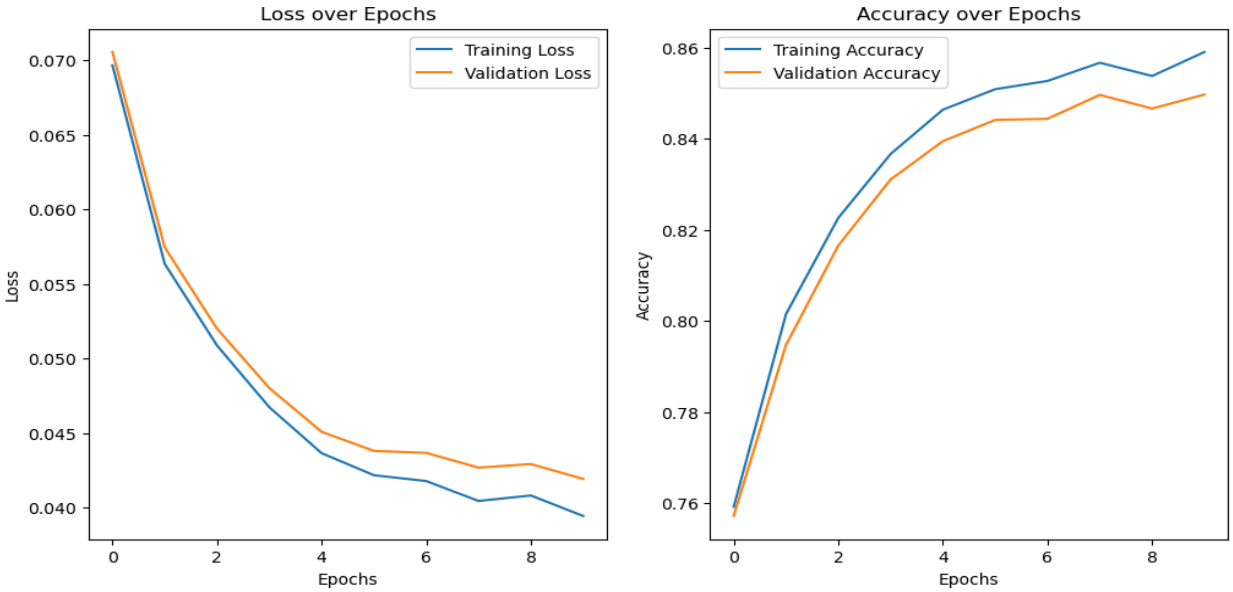
Epoch 7/10 - Loss: 0.0418, Accuracy: 0.8527, Val Loss: 0.0437, Val Accuracy: 0.8444, Val F1: 0.8428

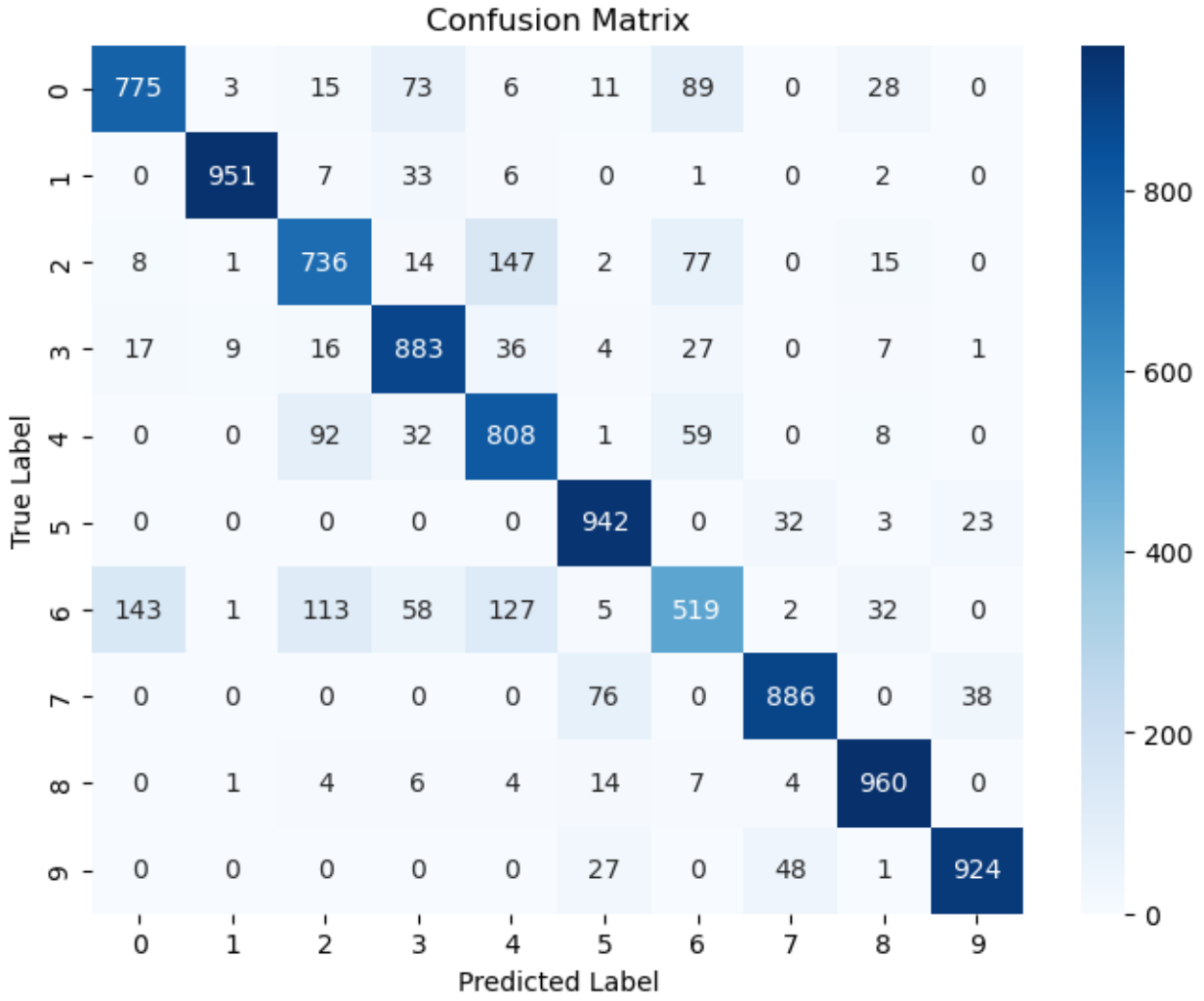
Epoch 8/10 - Loss: 0.0405, Accuracy: 0.8567, Val Loss: 0.0427, Val Accuracy: 0.8497, Val F1: 0.8477

Epoch 9/10 - Loss: 0.0408, Accuracy: 0.8538, Val Loss: 0.0429, Val Accuracy: 0.8467, Val F1: 0.8443

Epoch 10/10 - Loss: 0.0395, Accuracy: 0.8591, Val Loss: 0.0419, Val Accuracy: 0.8498, Val F1: 0.8480

Validation Loss: 0.0419, Validation Accuracy: 0.8498, Validation F1: 0.8480


Loss over Epochs
Training Loss
Validation Loss
Loss
Epochs
0.070
0.065
0.060
0.055
0.050
0.045
0.040
0
2
4
6
8
Accuracy over Epochs
Training Accuracy
Validation Accuracy
Accuracy
Epochs
0.86
0.84
0.82
0.80
0.78
0.76


Confusion Matrix
True Label
Predicted Label
0 775 3 15 73 6 11 89 0 28 0
1 0 951 7 33 6 0 1 0 2 0
2 8 1 736 14 147 2 77 0 15 0
3 17 9 16 883 36 4 27 0 7 1
4 0 0 92 32 808 1 59 0 8 0
5 0 0 0 0 0 942 0 32 3 23
6 143 1 113 58 127 5 519 2 32 0
7 0 0 0 0 0 76 0 886 0 38
8 0 1 4 6 4 14 7 4 960 0
9 0 0 0 0 0 27 0 48 1 924
0 1 2 3 4 5 6 7 8 9
800
600
400
200
0

Training model with architecture [784, 64, 10] and learning rate 0.0005

Epoch 1/10 - Loss: 0.0886, Accuracy: 0.7030, Val Loss: 0.0896, Val Accuracy: 0.7008, Val F1: 0.6852

Epoch 2/10 - Loss: 0.0698, Accuracy: 0.7433, Val Loss: 0.0708, Val Accuracy: 0.7400, Val F1: 0.7338

Epoch 3/10 - Loss: 0.0627, Accuracy: 0.7761, Val Loss: 0.0637, Val Accuracy: 0.7722, Val F1: 0.7673

Epoch 4/10 - Loss: 0.0576, Accuracy: 0.7929, Val Loss: 0.0588, Val Accuracy: 0.7900, Val F1: 0.7845

Epoch 5/10 - Loss: 0.0537, Accuracy: 0.8124, Val Loss: 0.0547, Val Accuracy: 0.8116, Val F1: 0.8071

Epoch 6/10 - Loss: 0.0525, Accuracy: 0.8158, Val Loss: 0.0537, Val Accuracy: 0.8122, Val F1: 0.8084

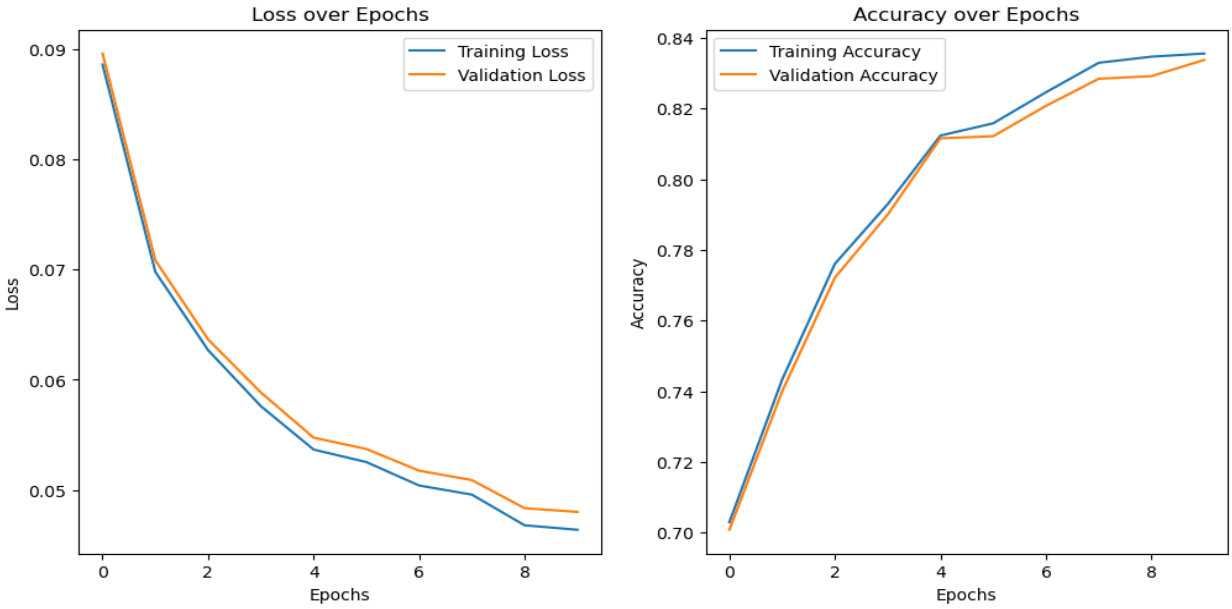
Epoch 7/10 - Loss: 0.0504, Accuracy: 0.8246, Val Loss: 0.0518, Val Accuracy: 0.8207, Val F1: 0.8177

Epoch 8/10 - Loss: 0.0496, Accuracy: 0.8329, Val Loss: 0.0509, Val Accuracy: 0.8284, Val F1: 0.8250

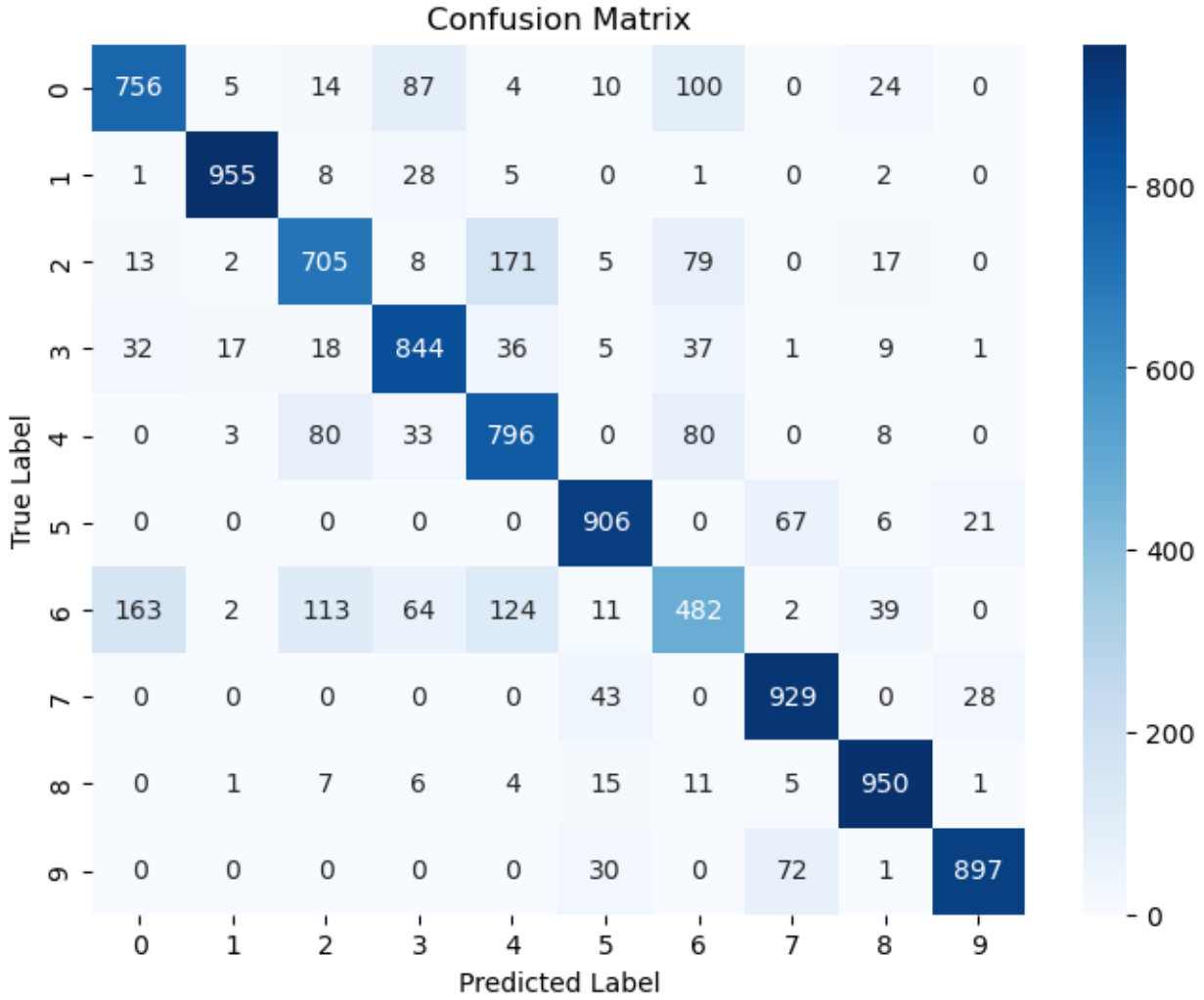
Epoch 9/10 - Loss: 0.0468, Accuracy: 0.8347, Val Loss: 0.0484, Val Accuracy: 0.8292, Val F1: 0.8251

Epoch 10/10 - Loss: 0.0464, Accuracy: 0.8356, Val Loss: 0.0480, Val Accuracy: 0.8337, Val F1: 0.8312

Validation Loss: 0.0480, Validation Accuracy: 0.8337, Validation F1: 0.8312


Loss over Epochs
Training Loss
Validation Loss
Loss
Epochs
0.09
0.08
0.07
0.06
0.05
0
2
4
6
8

Accuracy over Epochs
Training Accuracy
Validation Accuracy
Accuracy
Epochs
0.84
0.82
0.80
0.78
0.76
0.74
0.72
0.70
0
2
4
6
8


Confusion Matrix
True Label
Predicted Label
0 756 5 14 87 4 10 100 0 24 0
1 1 955 8 28 5 0 1 0 2 0
2 13 2 705 8 171 5 79 0 17 0
3 32 17 18 844 36 5 37 1 9 1
4 0 3 80 33 796 0 80 0 8 0
5 0 0 0 0 0 906 0 67 6 21
6 163 2 113 64 124 11 482 2 39 0
7 0 0 0 0 0 43 0 929 0 28
8 0 1 7 6 4 15 11 5 950 1
9 0 0 0 0 0 30 0 72 1 897
0 1 2 3 4 5 6 7 8 9
800
600
400
200
0

Training model with architecture [784, 128, 64, 10] and learning rate 0.005

Epoch 1/10 - Loss: 0.0879, Accuracy: 0.6215, Val Loss: 0.0886, Val Accuracy: 0.6244, Val F1: 0.5989

Epoch 2/10 - Loss: 0.0796, Accuracy: 0.6768, Val Loss: 0.0804, Val Accuracy: 0.6773, Val F1: 0.6718

Epoch 3/10 - Loss: 0.0724, Accuracy: 0.7048, Val Loss: 0.0731, Val Accuracy: 0.7057, Val F1: 0.6833

Epoch 4/10 - Loss: 0.0691, Accuracy: 0.7369, Val Loss: 0.0701, Val Accuracy: 0.7388, Val F1: 0.7328

Epoch 5/10 - Loss: 0.0659, Accuracy: 0.7593, Val Loss: 0.0668, Val Accuracy: 0.7562, Val F1: 0.7474

Epoch 6/10 - Loss: 0.0642, Accuracy: 0.7604, Val Loss: 0.0651, Val Accuracy: 0.7615, Val F1: 0.7580

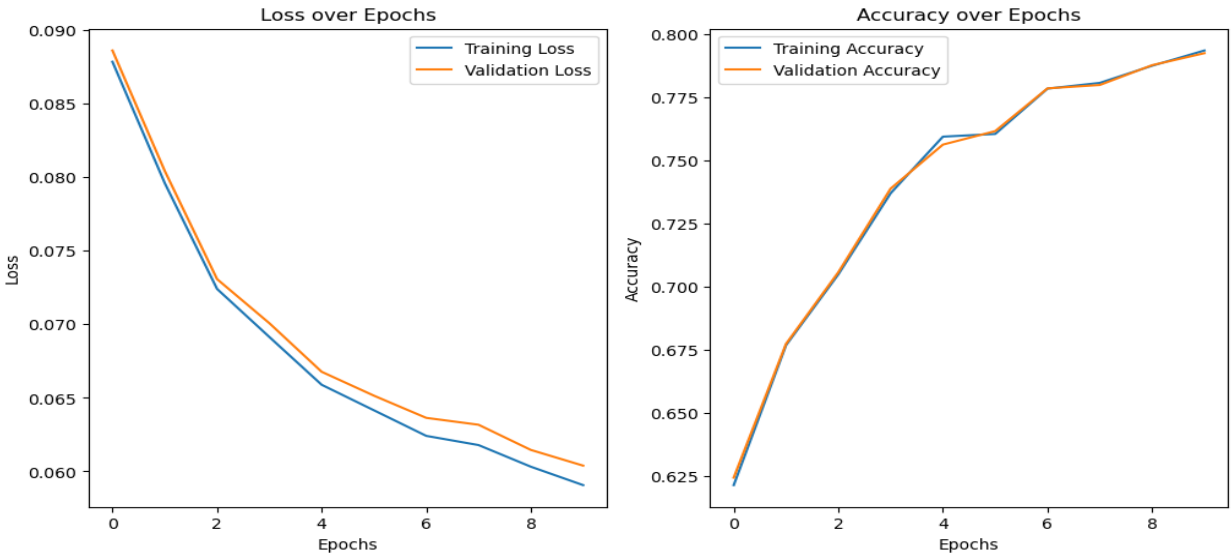
Epoch 7/10 - Loss: 0.0624, Accuracy: 0.7783, Val Loss: 0.0636, Val Accuracy: 0.7784, Val F1: 0.7774

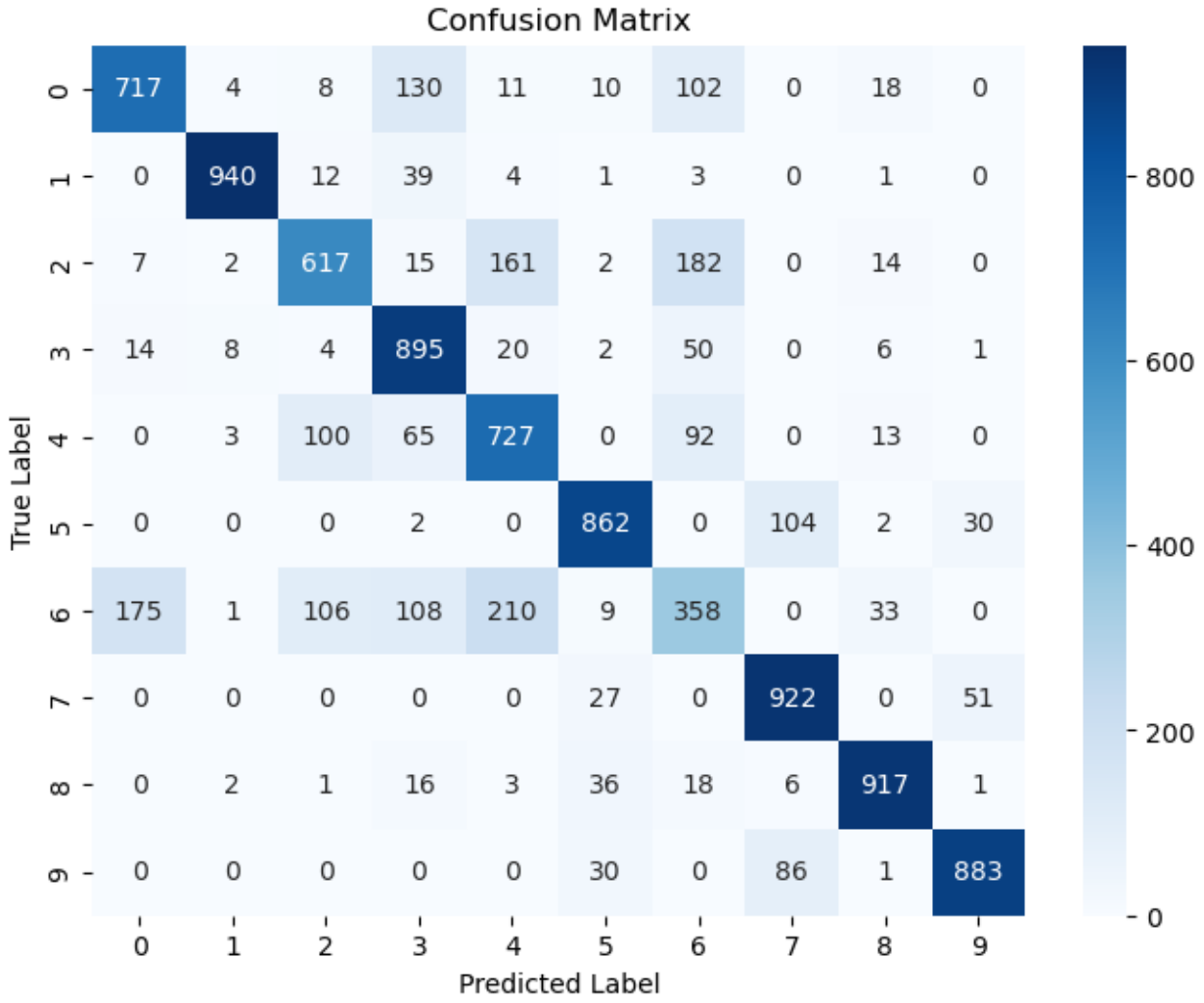
Epoch 8/10 - Loss: 0.0618, Accuracy: 0.7806, Val Loss: 0.0632, Val Accuracy: 0.7798, Val F1: 0.7771

Epoch 9/10 - Loss: 0.0603, Accuracy: 0.7874, Val Loss: 0.0615, Val Accuracy: 0.7876, Val F1: 0.7875

Epoch 10/10 - Loss: 0.0591, Accuracy: 0.7934, Val Loss: 0.0604, Val Accuracy: 0.7923, Val F1: 0.7899

Validation Loss: 0.0604, Validation Accuracy: 0.7923, Validation F1: 0.7899


Loss over Epochs
Training Loss
Validation Loss
Loss
Epochs
0.090
0.085
0.080
0.075
0.070
0.065
0.060
0
2
4
6
8
Accuracy over Epochs
Training Accuracy
Validation Accuracy
Accuracy
Epochs
0.800
0.775
0.750
0.725
0.700
0.675
0.650
0.625


Confusion Matrix
True Label
Predicted Label
0 717 4 8 130 11 10 102 0 18 0
1 0 940 12 39 4 1 3 0 1 0
2 7 2 617 15 161 2 182 0 14 0
3 14 8 4 895 20 2 50 0 6 1
4 0 3 100 65 727 0 92 0 13 0
5 0 0 0 2 0 862 0 104 2 30
6 175 1 106 108 210 9 358 0 33 0
7 0 0 0 0 0 27 0 922 0 51
8 0 2 1 16 3 36 18 6 917 1
9 0 0 0 0 0 30 0 86 1 883
0 1 2 3 4 5 6 7 8 9
800
600
400
200
0

Training model with architecture [784, 128, 64, 10] and learning rate 0.001

Epoch 1/10 - Loss: 0.0966, Accuracy: 0.6365, Val Loss: 0.0975, Val Accuracy: 0.6381, Val F1: 0.6295

Epoch 2/10 - Loss: 0.0786, Accuracy: 0.7046, Val Loss: 0.0796, Val Accuracy: 0.7053, Val F1: 0.7008

Epoch 3/10 - Loss: 0.0695, Accuracy: 0.7356, Val Loss: 0.0705, Val Accuracy: 0.7361, Val F1: 0.7255

Epoch 4/10 - Loss: 0.0642, Accuracy: 0.7589, Val Loss: 0.0655, Val Accuracy: 0.7498, Val F1: 0.7360

Epoch 5/10 - Loss: 0.0577, Accuracy: 0.7929, Val Loss: 0.0588, Val Accuracy: 0.7892, Val F1: 0.7853

Epoch 6/10 - Loss: 0.0547, Accuracy: 0.8066, Val Loss: 0.0559, Val Accuracy: 0.8033, Val F1: 0.7986

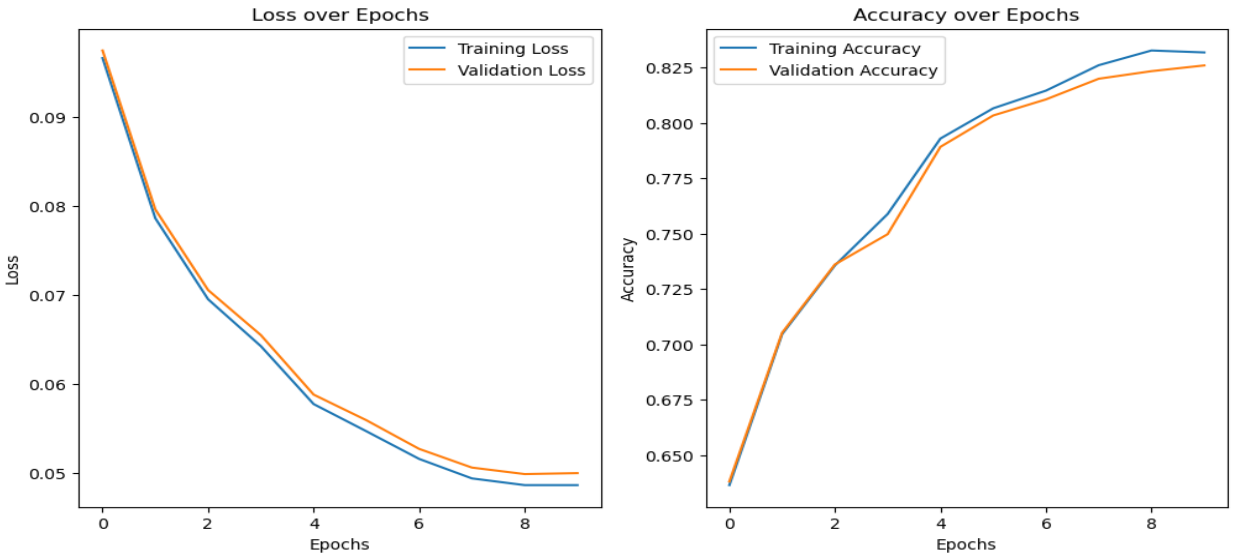
Epoch 7/10 - Loss: 0.0516, Accuracy: 0.8145, Val Loss: 0.0527, Val Accuracy: 0.8106, Val F1: 0.8061

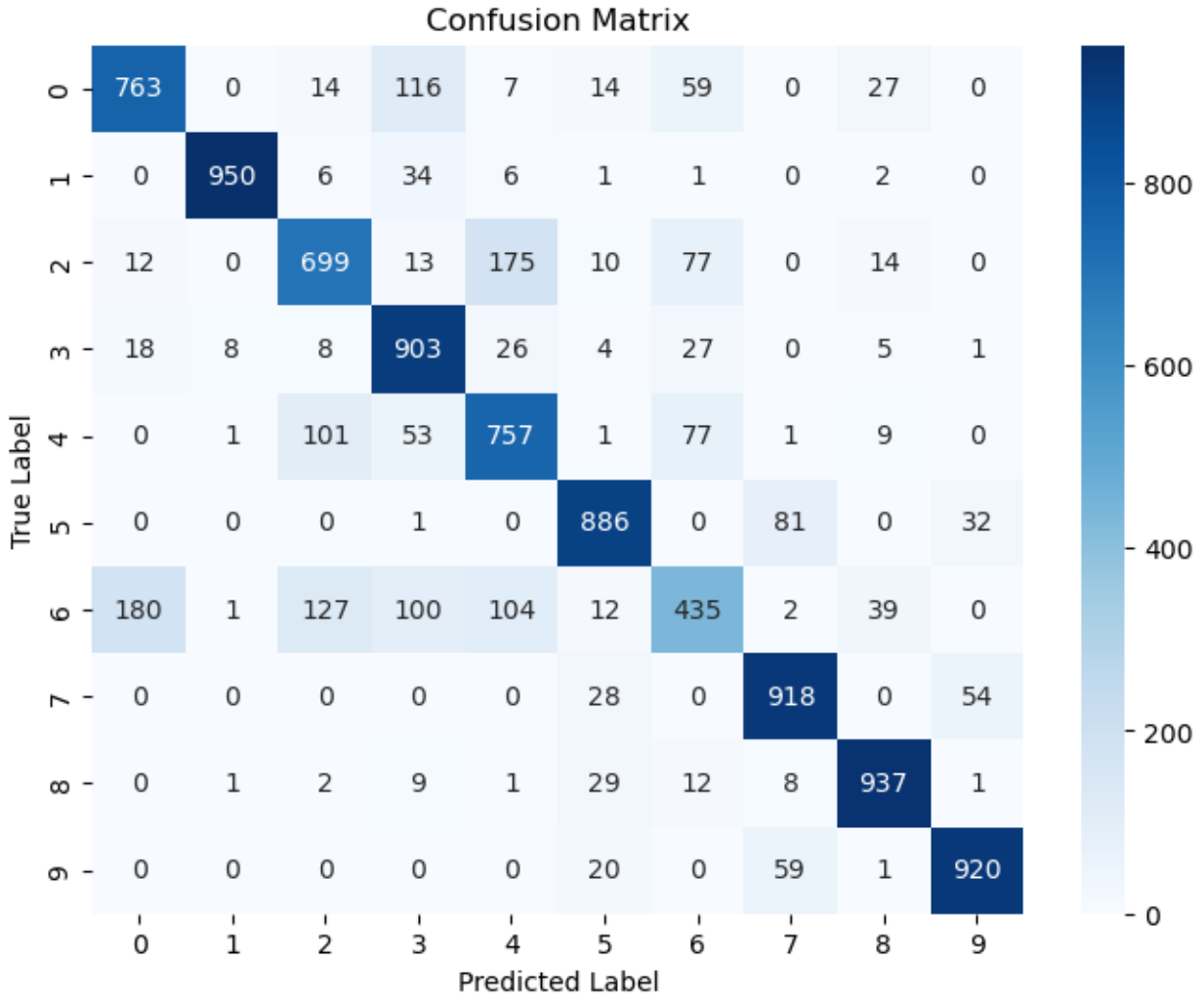
Epoch 8/10 - Loss: 0.0494, Accuracy: 0.8260, Val Loss: 0.0506, Val Accuracy: 0.8199, Val F1: 0.8161

Epoch 9/10 - Loss: 0.0486, Accuracy: 0.8327, Val Loss: 0.0499, Val Accuracy: 0.8233, Val F1: 0.8200

Epoch 10/10 - Loss: 0.0486, Accuracy: 0.8318, Val Loss: 0.0500, Val Accuracy: 0.8259, Val F1: 0.8222

Validation Loss: 0.0500, Validation Accuracy: 0.8259, Validation F1: 0.8222


Loss over Epochs
Training Loss
Validation Loss
Loss
Epochs
Accuracy over Epochs
Training Accuracy
Validation Accuracy
Accuracy
Epochs


Confusion Matrix
True Label
Predicted Label
0 763 0 14 116 7 14 59 0 27 0
1 0 950 6 34 6 1 1 0 2 0
2 12 0 699 13 175 10 77 0 14 0
3 18 8 8 903 26 4 27 0 5 1
4 0 1 101 53 757 1 77 1 9 0
5 0 0 0 1 0 886 0 81 0 32
6 180 1 127 100 104 12 435 2 39 0
7 0 0 0 0 0 28 0 918 0 54
8 0 1 2 9 1 29 12 8 937 1
9 0 0 0 0 0 20 0 59 1 920
0 1 2 3 4 5 6 7 8 9
800
600
400
200
0

Training model with architecture [784, 128, 64, 10] and learning rate 0.0005

Epoch 1/10 - Loss: 0.1239, Accuracy: 0.5572, Val Loss: 0.1244, Val Accuracy: 0.5587, Val F1: 0.5427

Epoch 2/10 - Loss: 0.1029, Accuracy: 0.6335, Val Loss: 0.1033, Val Accuracy: 0.6375, Val F1: 0.6256

Epoch 3/10 - Loss: 0.0817, Accuracy: 0.7084, Val Loss: 0.0825, Val Accuracy: 0.7078, Val F1: 0.6951

Epoch 4/10 - Loss: 0.0722, Accuracy: 0.7398, Val Loss: 0.0733, Val Accuracy: 0.7368, Val F1: 0.7223

Epoch 5/10 - Loss: 0.0683, Accuracy: 0.7498, Val Loss: 0.0692, Val Accuracy: 0.7468, Val F1: 0.7372

Epoch 6/10 - Loss: 0.0613, Accuracy: 0.7798, Val Loss: 0.0626, Val Accuracy: 0.7732, Val F1: 0.7691

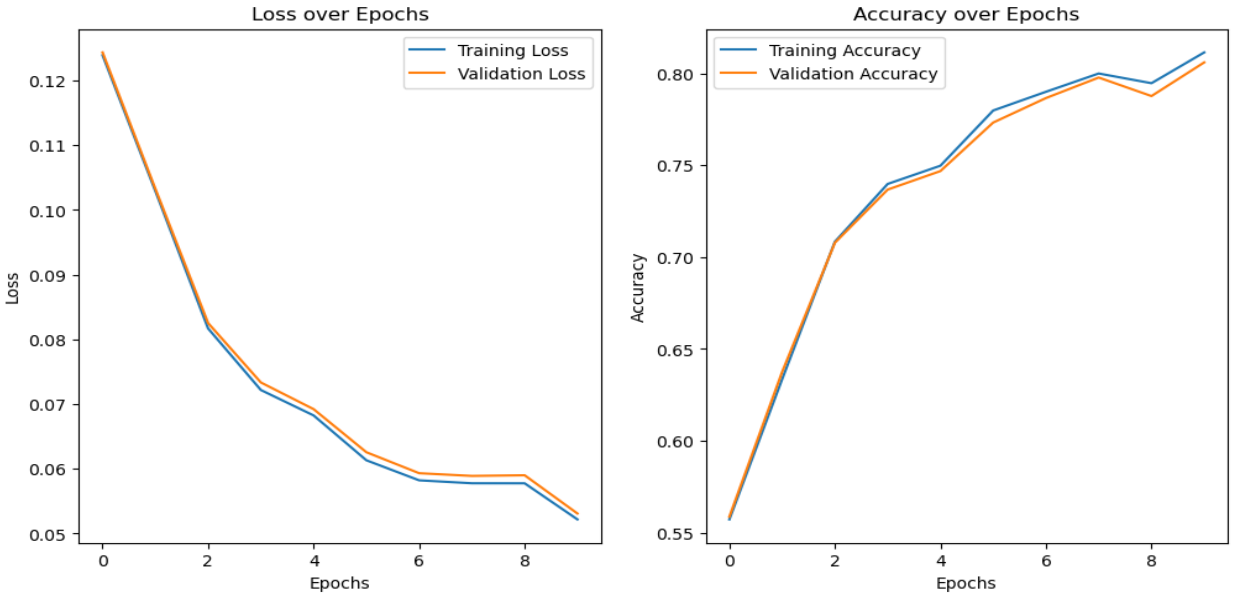
Epoch 7/10 - Loss: 0.0582, Accuracy: 0.7900, Val Loss: 0.0593, Val Accuracy: 0.7866, Val F1: 0.7833

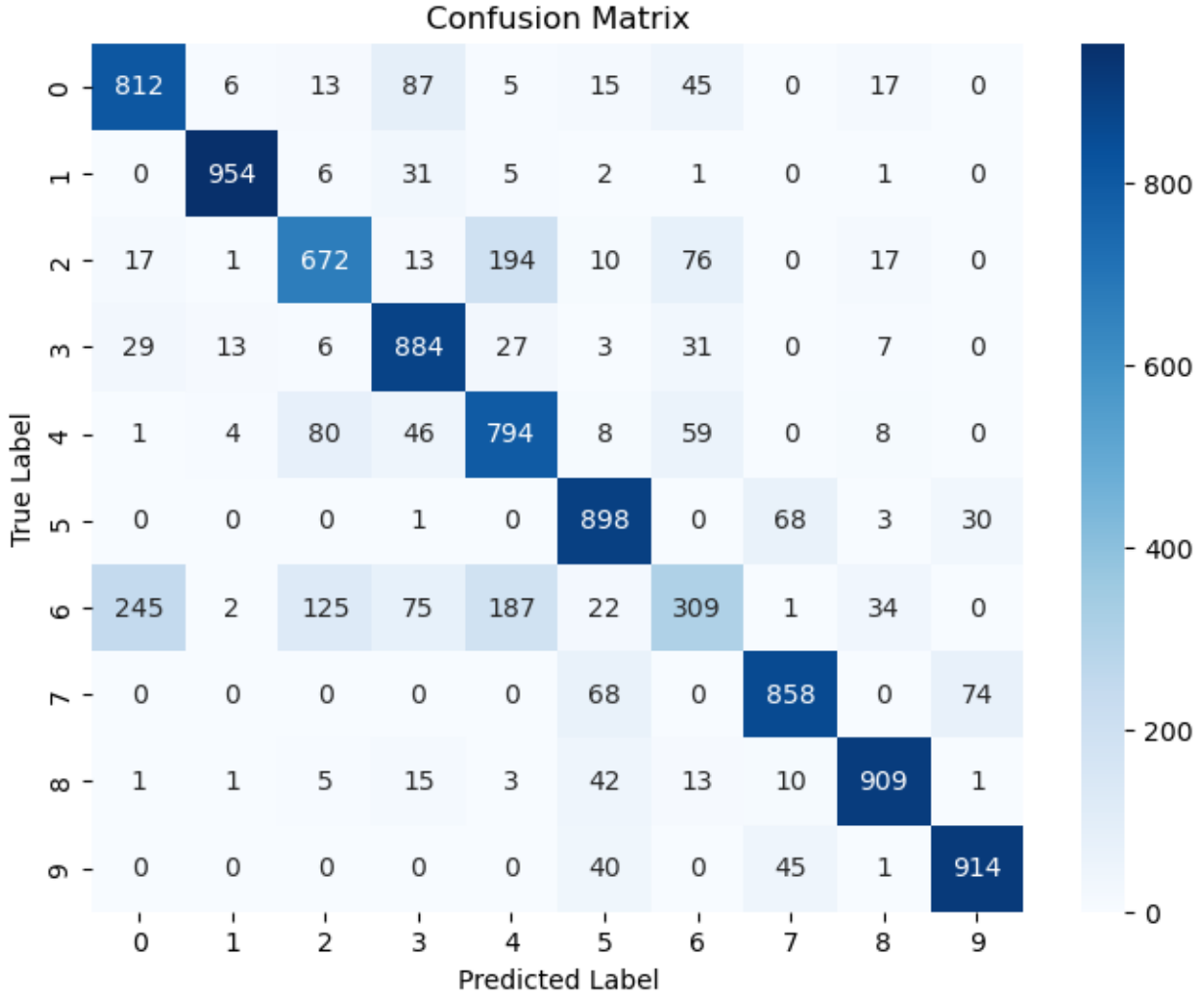
Epoch 8/10 - Loss: 0.0578, Accuracy: 0.8000, Val Loss: 0.0589, Val Accuracy: 0.7977, Val F1: 0.7934

Epoch 9/10 - Loss: 0.0578, Accuracy: 0.7947, Val Loss: 0.0590, Val Accuracy: 0.7877, Val F1: 0.7807

Epoch 10/10 - Loss: 0.0522, Accuracy: 0.8114, Val Loss: 0.0531, Val Accuracy: 0.8060, Val F1: 0.7988

Validation Loss: 0.0531, Validation Accuracy: 0.8060, Validation F1: 0.7988


Loss over Epochs
Training Loss
Validation Loss
Loss
Epochs
Accuracy over Epochs
Training Accuracy
Validation Accuracy
Accuracy
Epochs


Confusion Matrix
True Label
Predicted Label
812 6 13 87 5 15 45 0 17 0
0 954 6 31 5 2 1 0 1 0
17 1 672 13 194 10 76 0 17 0
29 13 6 884 27 3 31 0 7 0
1 4 80 46 794 8 59 0 8 0
0 0 0 1 0 898 0 68 3 30
245 2 125 75 187 22 309 1 34 0
0 0 0 0 0 68 0 858 0 74
1 1 5 15 3 42 13 10 909 1
0 0 0 0 0 40 0 45 1 914
0 1 2 3 4 5 6 7 8 9
800
600
400
200
0

Best Model Architecture: [784, 128, 10], Learning Rate: 0.005

Test Loss: 0.0399, Test Accuracy: 0.8576, Test F1: 0.8567

